Answer Key & Teacher Notes

Session 1 — Find the Whole · Session 2 — Solve for the Whole

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages: 8 problems, Find the Error, 2 test items. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Teaching Support Guide	Teacher page: lesson route, what to look for on each exit item, the next prompt, and a fresh check for after reteaching.
Explain Your Thinking	Optional language support for the Guided Notes "Model, Reason, Explain" page: draw it, talk it, write it.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank (English and Spanish), step frames on every problem and larger type (12 pt).

SESSION 1 · TEACHER NOTES

Find the Whole

TALK PROMPTS

- Why does finding 10% first make the whole easier to build?
- Before you scale up, how do you know the total has to be greater than the part?

PARTNER TASK

Puzzle Pieces and Water Bottle. One partner steps down to 10% (or 5%); the other builds up to 100% and checks with the percent. Switch roles with Library Books on the ratio table: 48 books is 60% (80 books).

WATCH FOR

- Multiplying the part by the percent (0.8 × 36) instead of scaling up, giving a whole smaller than the part.
- Dividing by the percent as a whole number (36 ÷ 80) instead of stepping down to 10%.
- Forgetting that the whole is 100% and stopping at the 10% amount.

SESSION 2 · TEACHER NOTES

Solve for the Whole

TALK PROMPTS

- Why does the unknown whole have to be larger than the part in a problem like this one?
- Why does dividing 312 by 0.65 give a number larger than 312?

PARTNER TASK

Fundraiser. A team raised $312, which is 65% of its goal. One partner writes the equation 0.65 × w = 312; the other solves (w = $480) and checks forward. Switch roles with Concert Ticket.

WATCH FOR

- Multiplying the part by the percent (w = 0.25 × 52) instead of dividing.
- Dividing by the percent number (1,050 ÷ 35) instead of the decimal (1,050 ÷ 0.35).
- Putting the known amount in place of the whole in the equation (w = 0.75 × 27).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S1 · Form B	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S2 · Form A	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S2 · Form B	6.AT.4	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

part: The amount you **know**; it matches a percent.

whole: The full amount. It is **100%**.

double number line: Two lines that **match** amounts with percents.

- The whole is 100%. When the percent is less than 100%, the whole is **greater** than the part.
- Divide to find a small percent like **10%**, then multiply up to 100%.
- Check: the percent of your whole gives back the **part**.

Example: 40% ÷ 4 = 10%, so 24 ÷ 4 = **6** stickers. 10% × 10 = 100%, so 6 × 10 = **60** stickers. Check: 0.40 × 60 = **24** Answer: The pack has **60** stickers.

YOU TRY

1. a. 80 b. 60 c. 50

2. 25% = 6; students: 6, 12, 18, 24. The class has 24 students.

EXIT TICKET

1. a. 110 b. 36

2. 10% = 28 ÷ 4 = 7; 100% = 70 games

3. **C** (greater than the part)

PRACTICE FORM A

1. a. 90 b. 80 c. 70 d. 48 e. 50 f. 40

2. 10% = $54 ÷ 3 = $18; dollars: 18, 36, 54, 72, 90, 108, 126, 144, 162, 180. Full cost = $180. Check: 0.30 × 180 = 54.

3. a. 84 ÷ 7 = 12 b. 12 × 10 = 120 pieces

4. a. 36 ÷ 3 = 12 mL b. 12 × 20 = 240 mL

5. 10%: 48 ÷ 6 = 8 books; 100%: 8 × 10 = 80 books

6. No. 10% = $21 ÷ 7 = $3, so 100% = $30. Kip multiplied 0.7 × 21; the first price must be more than $21.

7. 25% = 96 ÷ 3 = 32; seats: 32, 64, 96, 128. There are 128 seats.

8. a. 4% = 84 ÷ 7 = 12; 100% = 12 × 25 = 300 books b. 300 − 84 = 216 books c. 0.28 × 300 = 84, which matches Friday

9. The first price must be more than the $36 sale price, so multiplying by 0.8 is the wrong move. 10% = 36 ÷ 8 = $4.50, so 100% = $45. Check: 0.8 × 45 = 36.

10. **B** (60)

11. **C** (56)

Self-check word **SCALING**

PRACTICE FORM B

1. a. 70 b. 90 c. 84 d. 60 e. 80 f. 30

2. 10% = $42 ÷ 3 = $14; dollars: 14, 28, 42, 56, 70, 84, 98, 112, 126, 140. Full cost = $140. Check: 0.30 × 140 = 42.

3. a. 63 ÷ 7 = 9 b. 9 × 10 = 90 pieces

4. a. 27 ÷ 3 = 9 mL b. 9 × 20 = 180 mL

5. 10%: 42 ÷ 6 = 7 books; 100%: 7 × 10 = 70 books

6. No. 10% = $28 ÷ 7 = $4, so 100% = $40. Zara multiplied 0.7 × 28; the first price must be more than $28.

7. 25% = 108 ÷ 3 = 36; seats: 36, 72, 108, 144. There are 144 seats.

8. a. 4% = 72 ÷ 6 = 12; 100% = 12 × 25 = 300 books b. 300 − 72 = 228 books c. 0.24 × 300 = 72, which matches Monday

9. The first price must be more than the $45 sale price, so multiplying by 0.9 is the wrong move. 10% = 45 ÷ 9 = $5, so 100% = $50. Check: 0.9 × 50 = 45.

10. **B** (60)

11. **C** (50)

Self-check word **PERCENT**

GUIDED NOTES

variable: A letter that stands for an **unknown** number.

equation: A math sentence with an **equal** sign.

inverse operation: An operation that **undoes** another. Division undoes multiplication.

- Write the equation percent × whole = part, with the percent as a **decimal**.
- Divide the part by the decimal to find ***w***.
- Check: multiply your *w* by the percent to get the **part**.

Example: 35% = **0.35** Equation: 0.35 × *w* = **42** *w* = 42 ÷ 0.35 = **120** Answer: The team scored **120** points.

YOU TRY

1. a. 46 b. 45 c. 50

2. a. 0.15 × w = 7.20 b. w = 7.20 ÷ 0.15 = $48

EXIT TICKET

1. a. 52 b. 70

2. a. 0.6 × w = 138 b. w = 138 ÷ 0.6 = $230

3. **B** (40)

PRACTICE FORM A

1. a. 80 b. 60 c. 76 d. 68 e. 70 f. 65

2. a. 0.7 × w = 18,900 b. w = 18,900 ÷ 0.7 = 27,000; they paid $27,000

3. 0.4 × w = 84; w = 84 ÷ 0.4 = $210

4. a. 0.8 × w = 96, so w = 120 minutes b. 120 − 96 = 24 minutes

5. 0.65 × w = 312; w = 312 ÷ 0.65 = $480

6. No. The unknown whole goes with the percent: 0.75 × w = 27, so w = 27 ÷ 0.75 = $36.

7. 0.35 × w = 1,050; w = $3,000. Left after rent: $3,000 − $1,050 = $1,950.

8. a. 0.7 × j = 63, j = $90; 0.45 × b = 63, b = $140 b. the boots: $140 − $63 = $77 off, compared with $90 − $63 = $27 off c. $63 is a smaller percent of the boots' price, so the boots' first price had to be greater

9. Her whole came out smaller than the part. $52 is the part, so the equation is 0.25 × w = 52, and w = 52 ÷ 0.25 = $208. Check: 0.25 × 208 = 52.

10. **B** (0.2 × *w* = 30)

11. **C** (60)

Self-check word **DIVIDE**

PRACTICE FORM B

1. a. 70 b. 48 c. 68 d. 82 e. 60 f. 55

2. a. 0.6 × w = 16,800 b. w = 16,800 ÷ 0.6 = 28,000; they paid $28,000

3. 0.4 × w = 66; w = 66 ÷ 0.4 = $165

4. a. 0.8 × w = 72, so w = 90 minutes b. 90 − 72 = 18 minutes

5. 0.65 × w = 273; w = 273 ÷ 0.65 = $420

6. No. The unknown whole goes with the percent: 0.75 × w = 24, so w = 24 ÷ 0.75 = $32.

7. 0.35 × w = 1,120; w = $3,200. Left after rent: $3,200 − $1,120 = $2,080.

8. a. 0.8 × j = 72, j = $90; 0.45 × b = 72, b = $160 b. the boots: $160 − $72 = $88 off, compared with $90 − $72 = $18 off c. $72 is a smaller percent of the boots' price, so the boots' first price had to be greater

9. She multiplied instead of dividing, so her whole is smaller than the part. w = 70 ÷ 0.35 = 200. Check: 0.35 × 200 = 70.

10. **B** (0.3 × *w* = 18)

11. **C** (60)

Self-check word **EQUALS**

SHEET 1 · FIND THE WHOLE

1. 20% = $18; dollars: 18, 36, 54, 72, 90. The helmet costs $90.
2. a. 130 b. 44 c. 70 d. 80 e. 140 f. 90
3. One part (25%) = 18 ÷ 3 = 6 gal; full = 4 × 6 = 24 gallons.
4. 25% = 120 ÷ 3 = 40 pages, so 100% = 4 × 40 = 160 pages.
5. A is greater: 20 is a smaller percent (40%) of A, so A must be bigger. A = 50 (10% = 5) and B = 25 (10% = 2.5, or 20 ÷ 0.8).

SHEET 2 · SOLVE FOR THE WHOLE

1. a. 70 b. 48 c. 80
2. a. 0.55 × w = 33 b. w = 33 ÷ 0.55 = 60
3. 0.75 × w = 315, w = 315 ÷ 0.75 = $420. Still needed: $420 − $315 = $105.
4. 0.35 × w = 42, w = 42 ÷ 0.35 = 4,200 ÷ 35 = $120. Check: 0.35 × 120 = 42.
5. He divided by 35 instead of 0.35 (35% = 0.35); a whole of 0.8 is smaller than the part, which makes no sense. w = 28 ÷ 0.35 = 80. Check: 0.35 × 80 = 28.
6. **B** (0.3 × *w* = 45)
7. **C** (30)

SESSION 1 · FIND THE WHOLE

a. 30% is 3 parts of 10%, so 10% = $27 ÷ 3 = $9. b. 100% is 10 parts: $9 × 10 = $90. c. One part of the bar is $9, and the whole bar is 10 parts, so the whole is $90. Check: 0.30 × 90 = 27. The whole is greater than $27 because 30% is less than 100%.

Challenge $8.10 is less than the $27 the club already has, but the whole must be greater than the part. Sam multiplied when he should divide: 27 ÷ 0.3 = 90, so the club needs $90.

SESSION 2 · SOLVE FOR THE WHOLE

a. 0.45 × w = 9 b. w = 9 ÷ 0.45 = 900 ÷ 45 = 20 km c. Check: 0.45 × 20 = 9. The whole race (20 km) is greater than the 9 km part. 45% is a little less than half, so the race is a little more than 2 × 9 = 18 km.

Challenge 45 × w = 9 gives w = 0.2 km, which is less than the 9 km already run. 45% must be written as the decimal 0.45: 0.45 × w = 9, so w = 20 km.

Score each task 0–3: 1 for a model that fits the numbers, 1 for correct math with units, 1 for an explanation that matches the model.